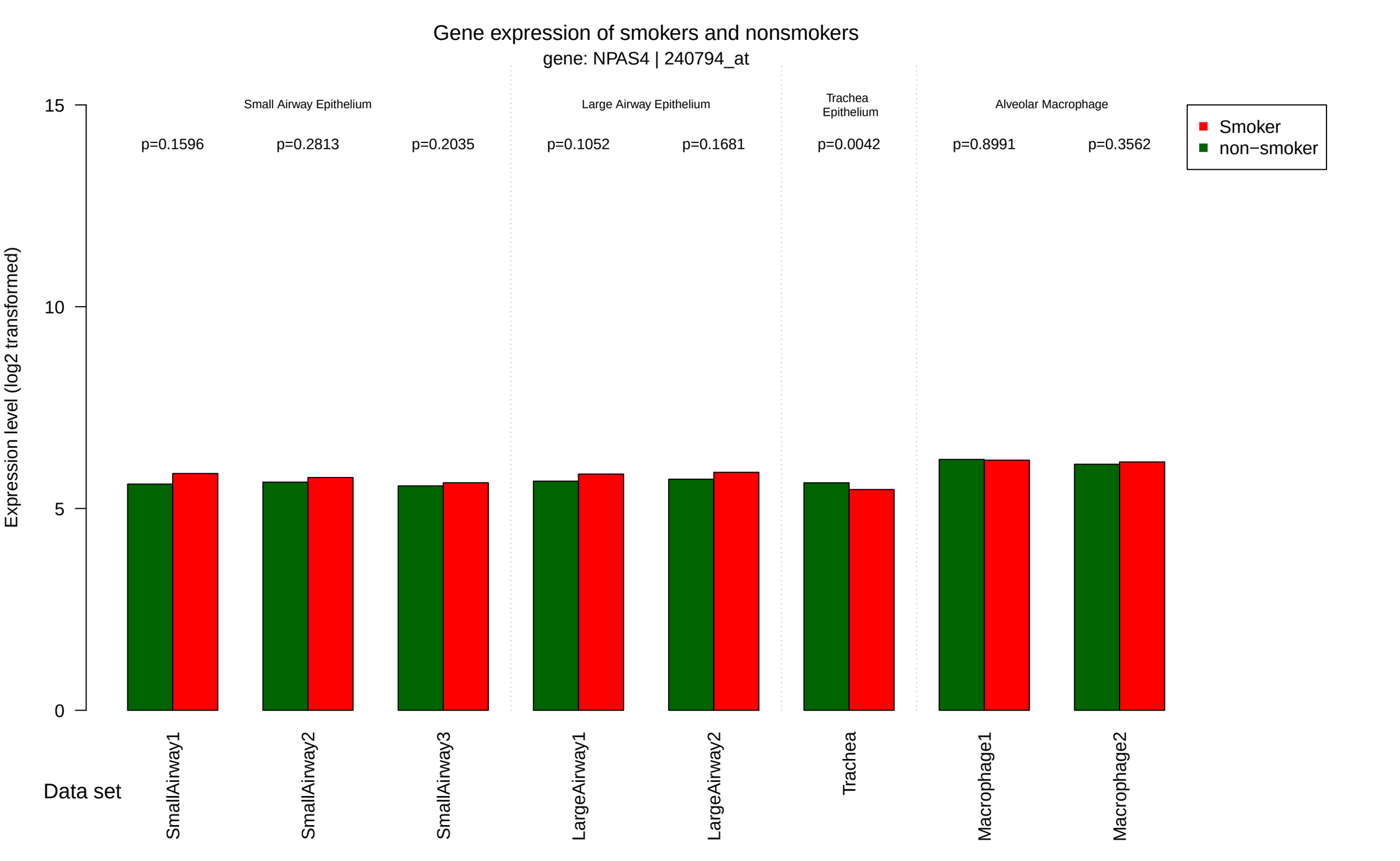
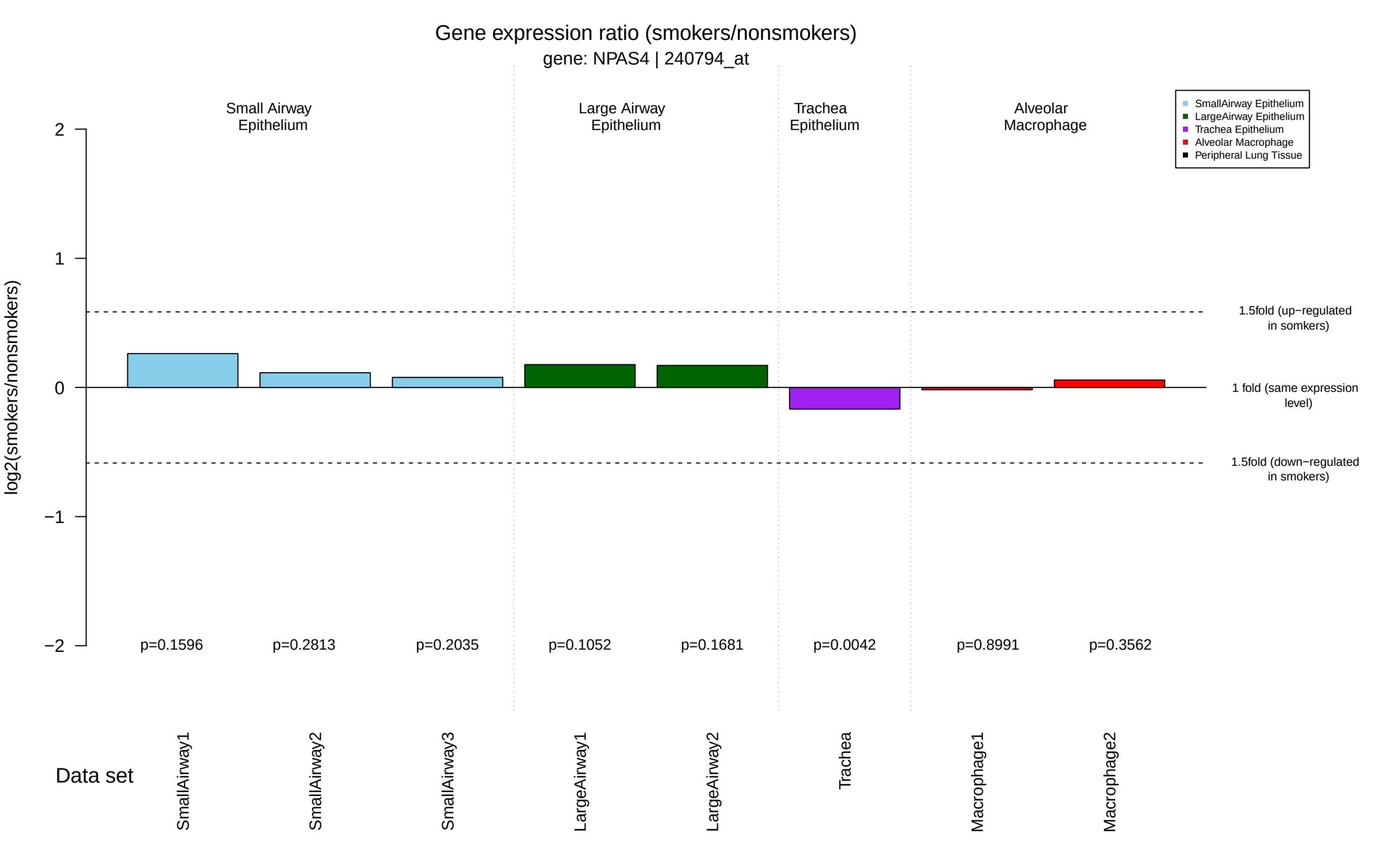
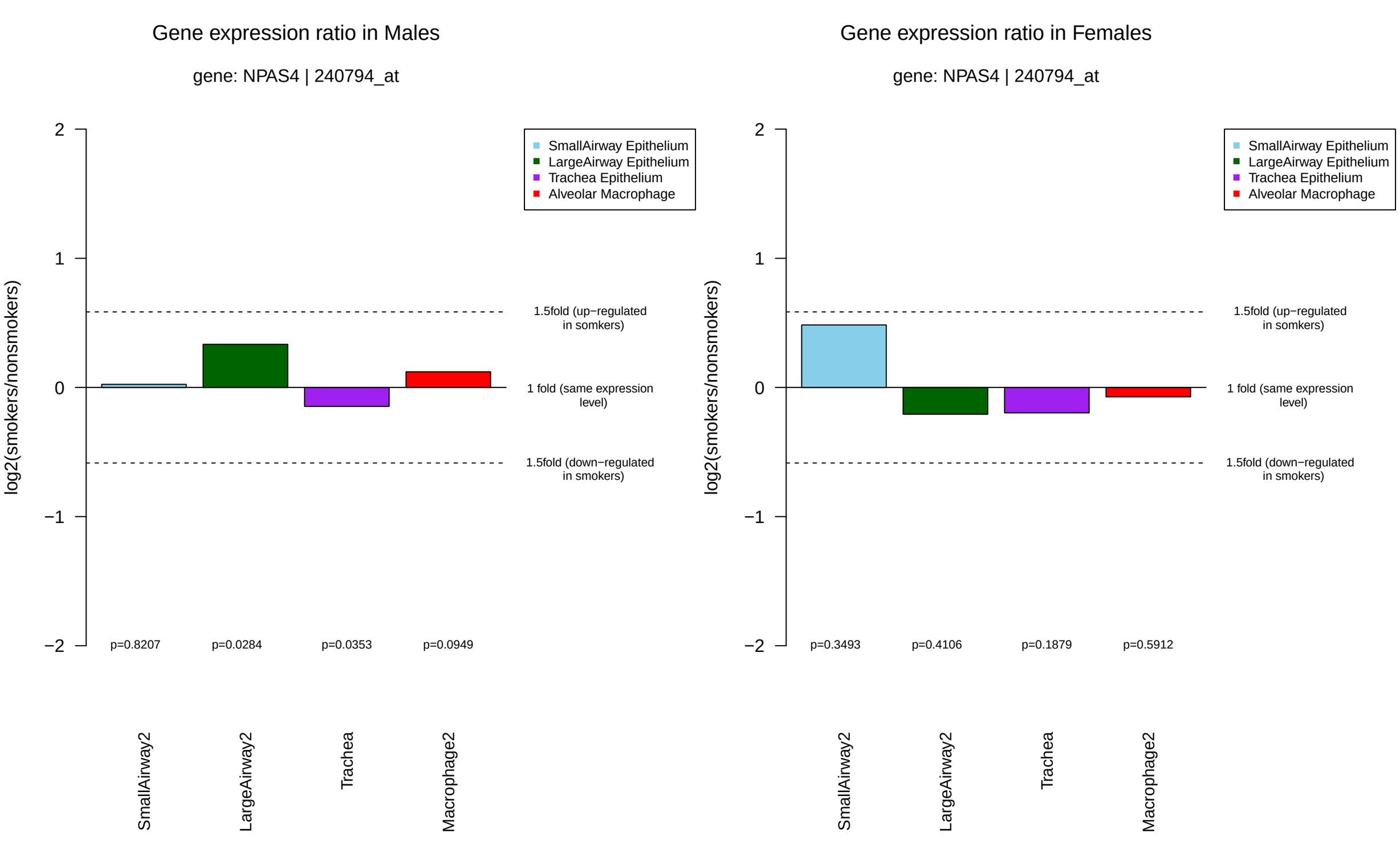
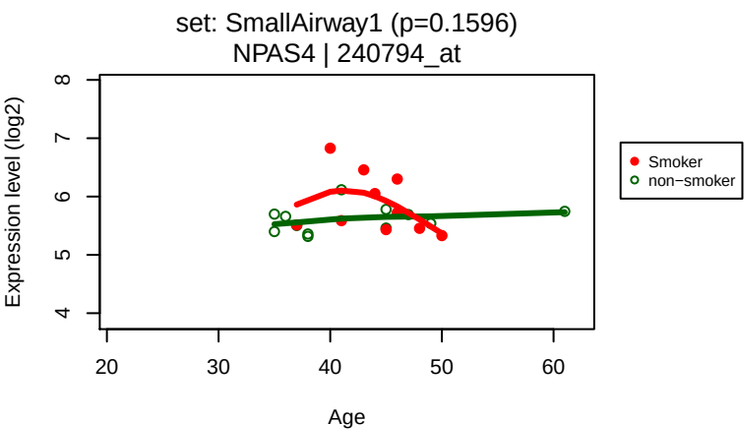
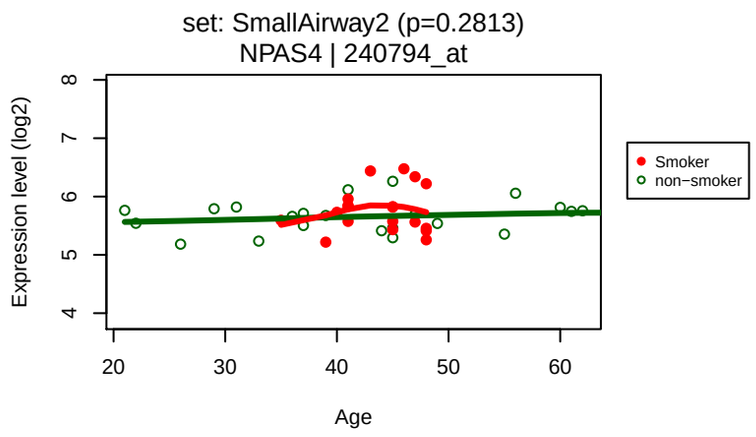
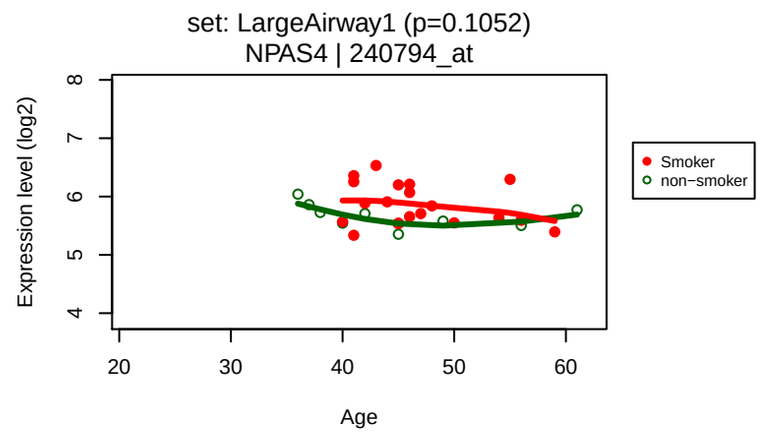

Gene expression of smokers and nonsmokers
gene: NPAS4 | 240794_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1596
p=0.2813
p=0.2035
p=0.1052
p=0.1681
p=0.0042
p=0.8991
p=0.3562
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

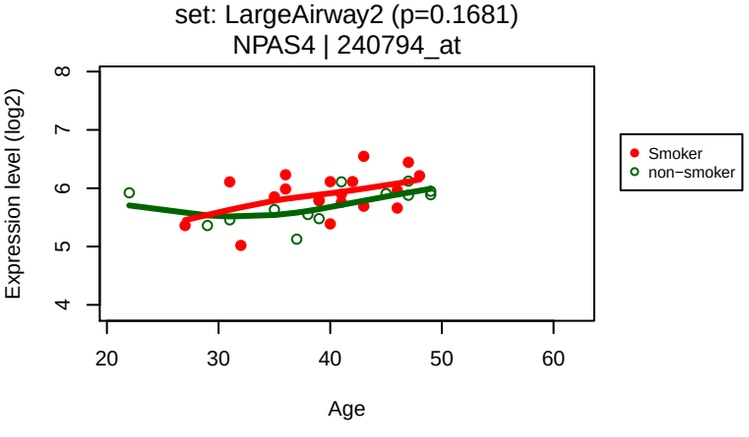

Gene expression ratio (smokers/nonsmokers)
gene: NPAS4 | 240794_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1596
p=0.2813
p=0.2035
p=0.1052
p=0.1681
p=0.0042
p=0.8991
p=0.3562
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

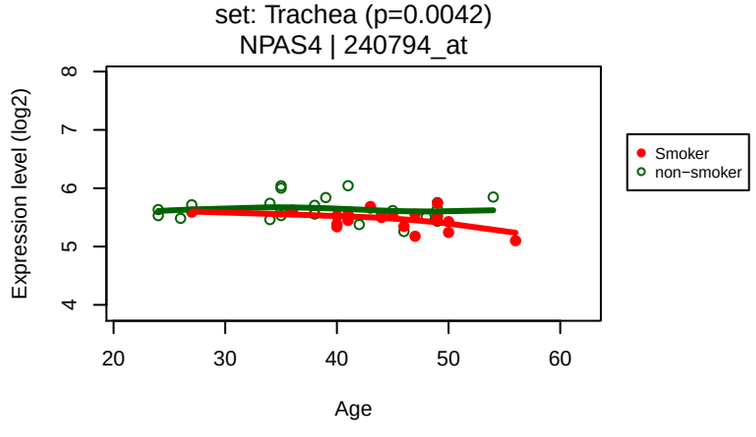

Gene expression ratio in Males
gene: NPAS4 | 240794_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8207
p=0.0284
p=0.0353
p=0.0949
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: NPAS4 | 240794_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3493
p=0.4106
p=0.1879
p=0.5912
SmallAirway2
LargeAirway2
Trachea
Macrophage2

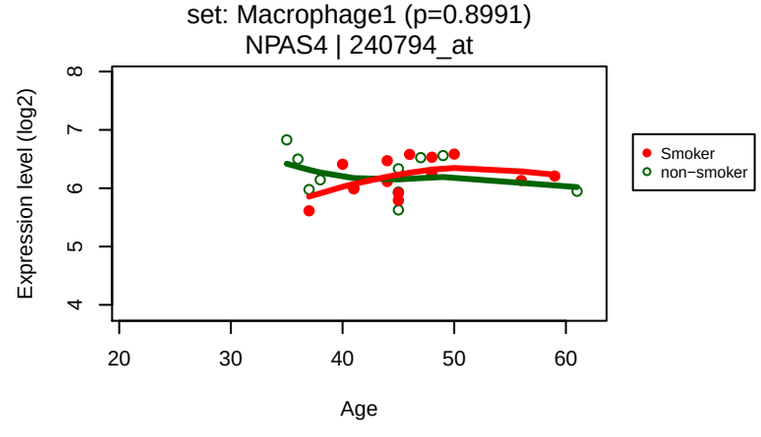

set: SmallAirway1 (p=0.1596)
NPAS4 | 240794_at
Expression level (log2)
Age
Smoker
non−smoker

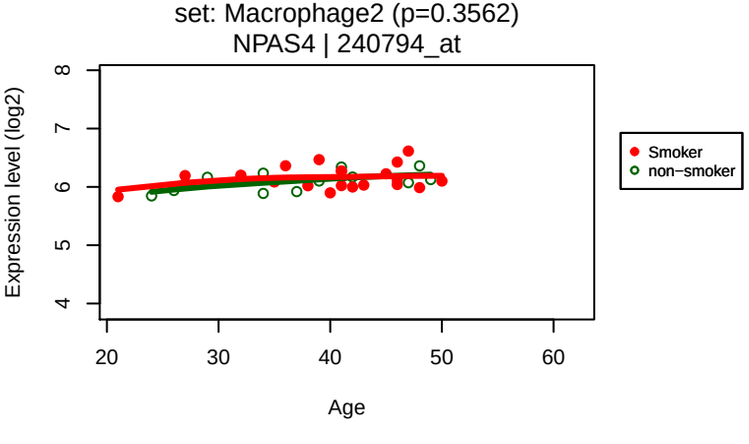

set: SmallAirway2 (p=0.2813)
NPAS4 | 240794_at
Expression level (log2)
Age
Smoker
non−smoker

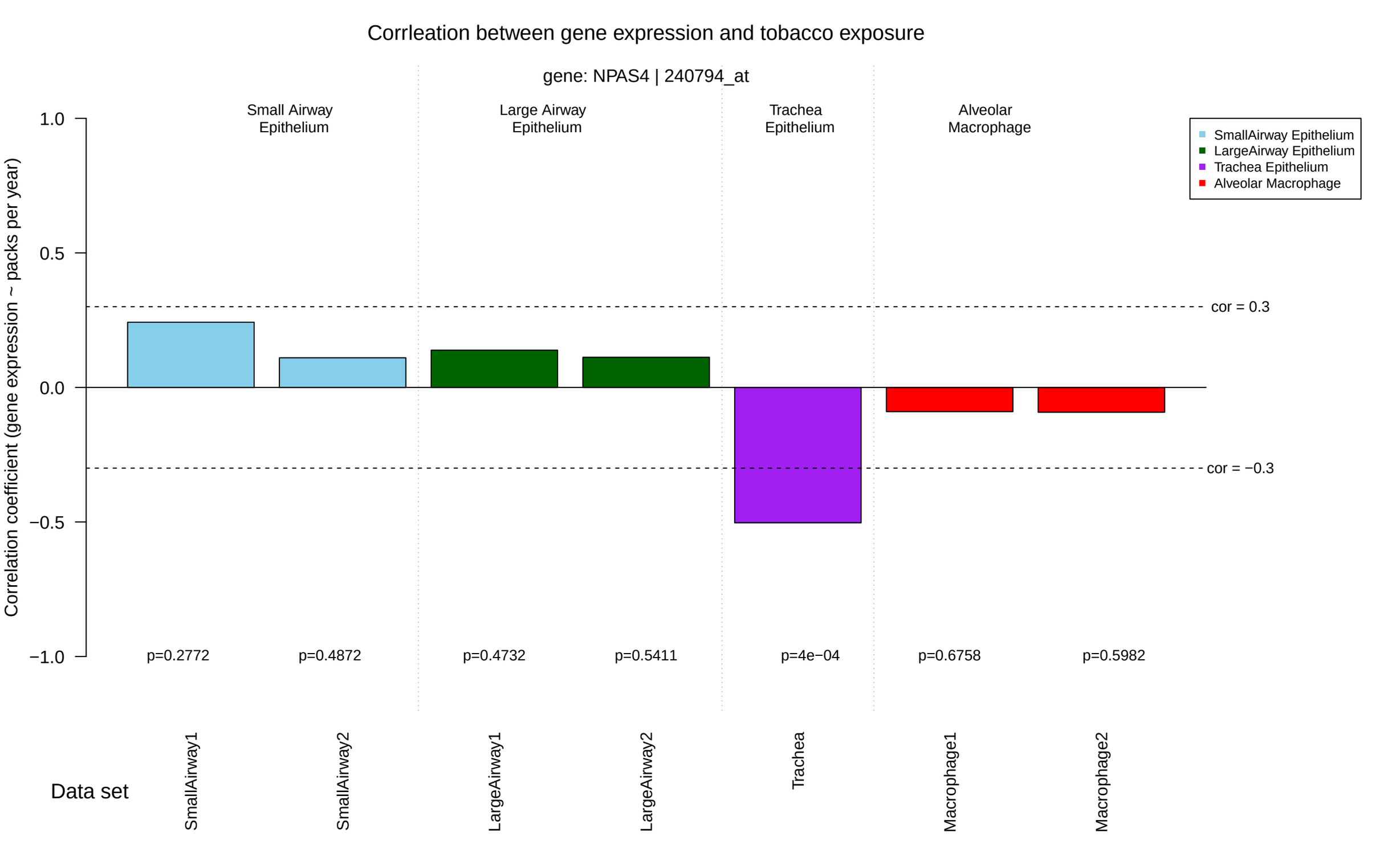

set: LargeAirway1 (p=0.1052)
NPAS4 | 240794_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1681)
NPAS4 | 240794_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0042)
NPAS4 | 240794_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8991)
NPAS4 | 240794_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3562)
NPAS4 | 240794_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: NPAS4 | 240794_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2772
p=0.4872
p=0.4732
p=0.5411
p=4e−04
p=0.6758
p=0.5982
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2